Protein Folding Simulation: Energy Landscapes and Folding Kinetics

Computational Structural Biology Laboratory

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Abstract

This computational study investigates protein folding through simplified lattice models and energy landscape analysis. We implement the hydrophobic-polar (HP) model to simulate folding on a 2D square lattice, employ Monte Carlo methods to explore conformational space, and analyze folding thermodynamics and kinetics. The simulations demonstrate spontaneous folding driven by hydrophobic collapse, visualize energy funnels characteristic of foldable sequences, and quantify structural convergence using RMSD and radius of gyration metrics. Results reproduce key principles of Anfinsen's thermodynamic hypothesis and provide insights into the folding free energy landscape.

1 Introduction

Protein folding represents one of the most fundamental problems in molecular biology: how does a linear amino acid sequence spontaneously adopt a unique three-dimensional structure? Anfinsen's thermodynamic hypothesis states that the native structure corresponds to the global minimum of Gibbs free energy under physiological conditions.

Definition 1.1 (Folding Free Energy) *The Gibbs free energy of folding is:*

$$\Delta G_{fold} = \Delta H_{fold} - T\Delta S_{fold} \tag{1}$$

where ΔH_{fold} represents enthalpic contributions (hydrogen bonds, van der Waals interactions, electrostatics) and $-T\Delta S_{fold}$ represents the entropic cost of constraining the unfolded ensemble.

2 Theoretical Framework

2.1 The HP Lattice Model

Definition 2.1 (HP Model) *The hydrophobic-polar (HP) model simplifies the 20 amino acids into two categories:*

- ***H*** *(hydrophobic): Ala, Val, Leu, Ile, Phe, Trp, Met*
- ***P*** *(polar): Ser, Thr, Asn, Gln, Asp, Glu, Lys, Arg, His, Tyr, Cys, Gly, Pro*

On a 2D square lattice, the energy is:

$$E = -\epsilon \sum_{\langle i,j \rangle} \delta_{H_i,H_j} \tag{2}$$

where the sum runs over non-bonded neighboring pairs and $\delta_{H_i,H_j} = 1$ *if both residues are hydrophobic (topological contacts).*

2.2 Energy Landscape Theory

Theorem 2.1 (Folding Funnel Hypothesis) *Foldable sequences exhibit a funnel-shaped energy landscape where many high-energy unfolded states progressively converge toward fewer low-energy native-like conformations. The funnel is characterized by:*

$$E(Q) = E_0 - \Delta E \cdot Q + k_B T \ln \Omega(Q) \tag{3}$$

where Q *is the fraction of native contacts,* $\Omega(Q)$ *is the density of states, and the landscape narrows as* $Q \to 1$*.*

2.3 Metropolis Monte Carlo

Definition 2.2 (Metropolis Criterion) *For a proposed conformational move* $i \to j$*, the acceptance probability is:*

$$P_{accept} = \min\left(1, \exp\left(-\frac{\Delta E}{k_B T}\right)\right) \tag{4}$$

where $\Delta E = E_j - E_i$*. This satisfies detailed balance and samples the canonical ensemble at temperature* T*.*

2.4 Structural Metrics

Definition 2.3 (Root Mean Square Deviation) *The RMSD between conformations* $\mathbf{r}$ *and* $\mathbf{r}_{ref}$ *after optimal superposition is:*

$$RMSD = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \|\mathbf{r}_i - \mathbf{r}_{ref,i}\|^2} \tag{5}$$

Definition 2.4 (Radius of Gyration) *The radius of gyration quantifies compactness:*

$$R_g = \sqrt{\frac{1}{N} \sum_{i=1}^{N} \|\mathbf{r}_i - \mathbf{r}_{cm}\|^2} \tag{6}$$

where $\mathbf{r}_{cm}$ *is the center of mass.*

3 Computational Simulation

4 Results

4.1 Thermodynamic Properties

4.2 Folding Kinetics

The folding trajectory at $T = 1.0$ demonstrates characteristic two-state behavior:

Example 4.1 (Two-State Folding) *At the optimal folding temperature (*$T \approx 1.0$*), the protein exhibits cooperative folding with rapid transition between extended and compact states. The energy barrier between these states is surmountable by thermal fluctuations, allowing efficient sampling of conformational space.*

5 Discussion

5.1 Energy Landscape Architecture

The simulations reveal a funnel-shaped energy landscape for the foldable sequence. At low temperatures ($T = 0.5$), the system rapidly descends to the global energy minimum corresponding to maximally compact conformations with hydrophobic residues in contact. At high temperatures ($T = 4.0$), thermal fluctuations dominate and the protein samples a broad ensemble of extended conformations.

Remark 5.1 (Levinthal's Paradox) *The folding funnel resolves Levinthal's paradox: proteins need not search all possible conformations exhaustively. Instead, the funnel topology provides a thermodynamic gradient that guides the search toward the native state, with many parallel pathways converging on similar low-energy structures.*

5.2 Hydrophobic Collapse

The radius of gyration analysis (Figure 1c) demonstrates that folding proceeds through rapid hydrophobic collapse:

This early collapse phase is followed by slower conformational rearrangements to optimize contact geometry.

5.3 Thermodynamic Transition

The heat capacity peak (Figure 1f) identifies the folding transition temperature T_f:

protein_folding_analysis.pdf

Figure 1: Comprehensive protein folding simulation using the HP lattice model: (a) Energy landscape showing energy-compactness correlation at different temperatures; (b) Energy trajectories demonstrating thermally-driven exploration and convergence; (c) Radius of gyration dynamics showing hydrophobic collapse at low temperature; (d) Final conformations at T=0.5, 1.0, 2.0, and 4.0 with hydrophobic residues (black) forming compact core; (e) Boltzmann energy distributions at equilibrium; (f) Heat capacity peak indicating folding transition temperature; (g) Native contact formation kinetics; (h) Free energy funnel representation; (i) Native state contact map showing non-local hydrophobic contacts.

At T_f, energy fluctuations are maximal because the system transitions between folded and unfolded ensembles. This is analogous to the latent heat of a first-order phase transition.

5.4 Comparison with Experimental Observables

While the HP model is highly simplified, it captures essential physics:

1. **Cooperativity**: Sharp transitions in structural properties
2. **Hydrophobic effect**: Driving force for collapse
3. **Funnel landscape**: Multiple pathways to native state
4. **Two-state kinetics**: Exponential approach to equilibrium

Remark 5.2 (Model Limitations) *The HP lattice model neglects:*

- *Specific side-chain interactions (hydrogen bonds, salt bridges)*
- *Backbone geometry and dihedral angle constraints*
- *Solvent effects beyond implicit hydrophobicity*
- *Chain flexibility variations*

More realistic models (MARTINI, AWSEM, Rosetta) incorporate these features at increased computational cost.

6 Conclusions

This computational study demonstrates key principles of protein folding thermodynamics and kinetics:

1. The HP lattice model successfully reproduces spontaneous folding driven by hydrophobic contacts, achieving final energies of **??** HP units at low temperature compared to **??** units at high temperature.
2. The folding transition occurs near $T_f \approx$ **??** with heat capacity maximum $C_V =$ **??**, indicating cooperative collapse.
3. Radius of gyration decreases from extended conformations ($R_g \approx$ **??**) to compact native-like states ($R_g \approx$ **??**), consistent with hydrophobic core formation.
4. Energy landscape analysis reveals funnel topology characteristic of foldable sequences, with convergence to low-energy states despite starting from random conformations.
5. Metropolis Monte Carlo efficiently samples conformational space, with acceptance rates decreasing at low temperature as the system becomes trapped near local minima.

Further Reading

- Dill, K.A., et al. “The Protein Folding Problem.” *Annual Review of Biophysics* 37 (2008): 289–316.
- Onuchic, J.N., et al. “Theory of Protein Folding: The Energy Landscape Perspective.” *Annual Review of Physical Chemistry* 48 (1997): 545–600.
- Bryngelson, J.D., and Wolynes, P.G. “Spin Glasses and the Statistical Mechanics of Protein Folding.” *Proceedings of the National Academy of Sciences* 84 (1987): 7524–7528.
- Lau, K.F., and Dill, K.A. “A Lattice Statistical Mechanics Model of the Conformational and Sequence Spaces of Proteins.” *Macromolecules* 22 (1989): 3986–3997.
- Fersht, A.R. “Nucleation Mechanisms in Protein Folding.” *Current Opinion in Structural Biology* 7 (1997): 3–9.
- Dobson, C.M., et al. “Protein Folding and Misfolding.” *Nature* 426 (2003): 884–890.
- Levinthal, C. “How to Fold Graciously.” *Mossbauer Spectroscopy in Biological Systems Proceedings* 67 (1969): 22–24.
- Anfinsen, C.B. “Principles that Govern the Folding of Protein Chains.” *Science* 181 (1973): 223–230.
- Karplus, M., and McCammon, J.A. “Molecular Dynamics Simulations of Biomolecules.” *Nature Structural Biology* 9 (2002): 646–652.
- Brooks, C.L., et al. “Taking a Walk on a Landscape.” *Science* 293 (2001): 612–613.
- Dill, K.A., and Chan, H.S. “From Levinthal to Pathways to Funnels.” *Nature Structural Biology* 4 (1997): 10–19.
- Wolynes, P.G., et al. “Navigating the Folding Routes.” *Science* 267 (1995): 1619–1620.
- Baker, D. “A Surprising Simplicity to Protein Folding.” *Nature* 405 (2000): 39–42.
- Frauenfelder, H., et al. “The Energy Landscapes and Motions of Proteins.” *Science* 254 (1991): 1598–1603.
- Leopold, P.E., et al. “Protein Folding Funnels: A Kinetic Approach to the Sequence-Structure Relationship.” *Proceedings of the National Academy of Sciences* 89 (1992): 8721–8725.

- Shakhnovich, E.I. "Protein Folding Thermodynamics and Dynamics: Where Physics, Chemistry, and Biology Meet." *Chemical Reviews* 106 (2006): 1559–1588.

- Jumper, J., et al. "Highly Accurate Protein Structure Prediction with AlphaFold." *Nature* 596 (2021): 583–589.

- Ferreiro, D.U., et al. "Localizing Frustration in Native Proteins and Protein Assemblies." *Proceedings of the National Academy of Sciences* 104 (2007): 19819–19824.

- Lindorff-Larsen, K., et al. "How Fast-Folding Proteins Fold." *Science* 334 (2011): 517–520.

- Thirumalai, D., et al. "Theoretical Perspectives on Protein Folding." *Annual Review of Biophysics* 39 (2010): 159–183.